Chapter 13 Differential Equations - Homework Answers

Chapter revision | Warm-up

1 **a** $3\log|x| + C$ **b** $\log(x^2+1) + C$ **c** $-\frac{1}{2}e^{-2x} + C$

2 **a** $y = \pm e^{3x+2}$, that is $y = Ae^{3x}$ with $A = \pm e^2$ **b** $\dfrac{y}{1-y} = e^x$, so $y = \dfrac{e^x}{1+e^x} = \dfrac{1}{1+e^{-x}}$

3 $\text{RHS} = \dfrac{1}{2} \times \dfrac{(2-x)+x}{x(2-x)} = \dfrac{1}{x(2-x)} = \text{LHS}$

4 $2x + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x}{y}$

5 $P(5) = 500e \approx 1359$; $e^{0.2t} = 4$ gives $t = 5\log 4 \approx 6.93$

Chapter 13 Differential Equations - Homework Answers

Lesson 13A | Differential equations and their solutions

Lesson Examples

C1 **Tutor demo. a** (i) order 3 (ii) not a DE, as there is no derivative (iii) order 1. **b** (i) $y' = 15e^{3x} = 3y$: a solution. (ii) $y' = e^x$ but $3y = 3e^x + 9$: not a solution.
Your turn. a (i) 2 (ii) 1. **b** LHS $= \sec^2 x$, RHS $= 1 + \tan^2 x = \sec^2 x$.

C2 **Tutor demo 1.** $y' = 2Ae^{2x}$ and $2y + 2 = 2Ae^{2x}$. $y(0) = 3$ gives $A = 4$: $y = 4e^{2x} - 1$.
Your turn 1. $y' = -Ce^{-x} + 1 = x - y$. Through $(0, 2)$: $C = 3$, so $y = 3e^{-x} + x - 1$.
Tutor demo 2. $y'' = -9A\cos 3x - 9B\sin 3x = -9y$. $y(0) = 2$ gives $A = 2$;
$y'(0) = 3B = 6$ gives $B = 2$: $y = 2\cos 3x + 2\sin 3x$.
Your turn 2. $y' = 3x^2 - 2$, $y = x^3 - 2x + 1$.

C3 **Tutor demo.** $2x + 2yy' = 0$, so $y' = -\dfrac{x}{y}$. The radius to (x, y) has gradient $\dfrac{y}{x}$, so the tangent is perpendicular to the radius.
Your turn. $y + xy' = 0$, so $y' = -\dfrac{y}{x}$. Yes: it is $y' + \dfrac{1}{x}y = 0$.

Textbook Homework

1 (i) Order: **a** first **b** first **c** second **d** first **e** second **f** first **g** first **h** second **i** second
(ii) Linear: **a** linear **b** non-linear **d** linear **f** non-linear **g** linear
(iii) Constants: **a** 1 **b** 1 **c** 2 **d** 1 **e** 2 **f** 1 **g** 1 **h** 2 **i** 2

3 **a** $y = x^2 - 3x + C$ **b** $y = -6e^{-2x} + 4x + C$ **c** $y = \tan x + C$ **d** $y = 3\sin 2x - 3\cos 3x + C$ **e** $y = \frac{2}{15}(1 + 5x)^{\frac{3}{2}} + C$ **f** $y = 2\sin x^2 + C$

6 **a** $y = x^2 + Ax + B$ **b** $y = -\frac{1}{4}\cos 2x + Ax + B$ **c** $y = 4e^{\frac{1}{2}x} + Ax + B$ **d** $y = -\log|\cos x| + Ax + B$

8 **a** $y = x - 1$ **b** $y = x^2 - 3x + 2$ **c** $y = x^3 + 3x^2 - 9x + 7$ **d** $y = 2 - \cos x$ **e** $y = 3(e^{2x} - 1)$ **f** $y = 2x\sqrt{x} - 2x - 1$

9 **b** $y = \log\left|\dfrac{x}{1-x}\right| + C$ **c** $y = \log\left|\dfrac{x}{1-x}\right|$

10 **a** (i) $2x + 2yy' = 0$ (iii) y' is undefined at $(3, 0)$ and $(-3, 0)$, where the tangent to the circle is vertical. **b** (i) $y' = \dfrac{1}{2y}$ (ii) $\dfrac{dy}{dx} = -\dfrac{y}{x}$ (iii) $\dfrac{dy}{dx} = -\dfrac{9x}{16y}$ (iv) $\dfrac{dy}{dx} = \dfrac{x}{4y}$

11 **b** When $y = 12$, $y'' = -y = -12$. **c** $f''(x)$ is the opposite of $f(x)$, so it is a reflection in the x-axis.

12 $y = 1 + x - \log(\cos x)$

13 **a** $\lambda = 1$ or 3 **b** $\lambda = -1$ **c** No real solution.

Chapter 13 Differential Equations - Homework Answers

Lesson 13A | Differential equations and their solutions

Past-Paper Homework - Worked Solutions

PP 1 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q7**

Substituting $y = e^{px}$: $p^2e^{px} - pe^{px} - 12e^{px} = 0$. Since $e^{px} \neq 0$, $p^2 - p - 12 = 0$, $(p-4)(p+3) = 0$, so $p = 4$ or $p = -3$. **Answer (B).**

PP 2 **Cherrybrook Technology High School, 2020, Year 12 AP4 Mathematics Extension 1 (Trial), Q8**

$a^2 + a - 6 = 0$, $(a+3)(a-2) = 0$, so $a = 2$ or $a = -3$. **Answer C.**

PP 3 **Sydney Technical High School, 2020, Trial Higher School Certificate Examination (Sample 1), Mathematics Extension 1, Q11(c)**

$y = x\ln x$, so $y' = \ln x + 1$ and $y'' = \dfrac{1}{x}$.

$$x^2y'' - xy' + y = x^2 \cdot \frac{1}{x} - x(\ln x + 1) + x\ln x = x - x\ln x - x + x\ln x = 0$$

PP 4 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(a)**

$y' = 3Ae^{3x} - 2Be^{-2x}$ and $y'' = 9Ae^{3x} + 4Be^{-2x}$.

$$y'' - y' - 6y = (9 - 3 - 6)Ae^{3x} + (4 + 2 - 6)Be^{-2x} = 0$$

for all real A and B.

PP 5 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q13(a)**

(i) Write $u = e^{-\frac{t}{3}}$. Then $K = \dfrac{21000}{7 + 3u}$ and $\dfrac{dK}{dt} = \dfrac{21000 \times 3u \times \frac{1}{3}}{(7+3u)^2} = \dfrac{21000u}{(7+3u)^2}$.

Also $1 - \dfrac{K}{3000} = 1 - \dfrac{7}{7+3u} = \dfrac{3u}{7+3u}$, so

$$\frac{1}{3}\left(1 - \frac{K}{3000}\right)K = \frac{1}{3} \times \frac{3u}{7+3u} \times \frac{21000}{7+3u} = \frac{21000u}{(7+3u)^2}. \checkmark$$

(ii) At $t = 0$: $K = \dfrac{21000}{10} = 2100$ kangaroos.

(iii) As $t \to \infty$, $e^{-\frac{t}{3}} \to 0$, so $K \to \dfrac{21000}{7} = 3000$ kangaroos.

(iv) $\dfrac{1}{K}\dfrac{dK}{dt} = \dfrac{1}{3}\left(1 - \dfrac{2100}{3000}\right) = \dfrac{1}{3} \times 0.3 = 0.1$, that is, 10% per year.

PP 6 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(b)**

(i) $\dfrac{dF}{dn} = -275ke^{-k(n-1)} = -k(F - F_0)$, since $F - F_0 = 275e^{-k(n-1)}$.

(ii) $n = 1$: $350 = 275 + F_0$, so $F_0 = 75$.

(iii) $80 = 275e^{-199k} + 75$, so $e^{-199k} = \dfrac{5}{275} = \dfrac{1}{55}$ and $k = \dfrac{\ln 55}{199}$.

(iv) As $n \to \infty$, $e^{-k(n-1)} \to 0$ (as $k > 0$), so $F \to 75$. The endurance limit is 75.

Chapter 13 Differential Equations - Homework Answers

Lesson 13B | Slope fields

Lesson Examples

C1 **Tutor demo.** Gradients $x - y$: at $x = -1$: $0, -1, -2$ for $y = -1, 0, 1$; at $x = 0$: $1, 0, -1$; at $x = 1$: $2, 1, 0$. The isoclines are the lines $y = x - c$, all with gradient 1.
Your turn. Gradients xy: at $x = -1$: $1, 0, -1$; at $x = 0$: $0, 0, 0$; at $x = 1$: $-1, 0, 1$. The gradient is zero on both axes.

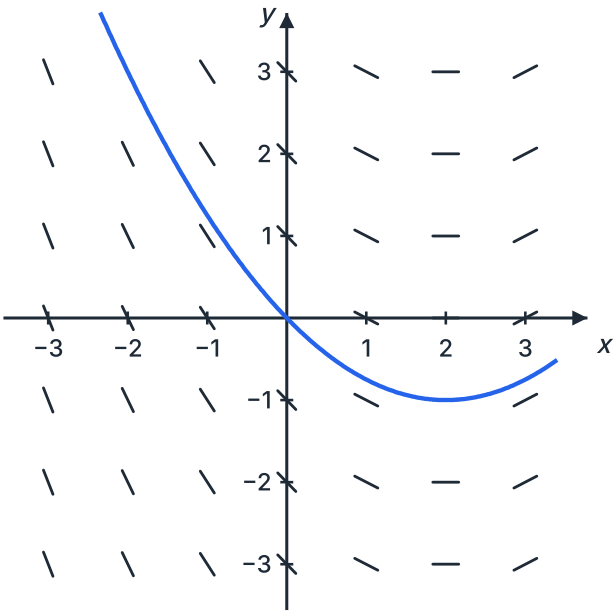
C2 **Tutor demo.** A, $y' = -xy$. D is excluded because its zero slopes lie on $y = -x$, not the y-axis. B has negative slopes in quadrant 2, and C has positive slopes in quadrant 3.

C3 **Your turn.** The constant solutions are $y = 0$ and $y = 3$. $y' > 0$ for $0 < y < 3$, and $y' < 0$ for $y > 3$ and for $y < 0$. Curves approach $y = 3$ from both sides as x increases, so $y = 3$ is stable and $y = 0$ is unstable.

Textbook Homework

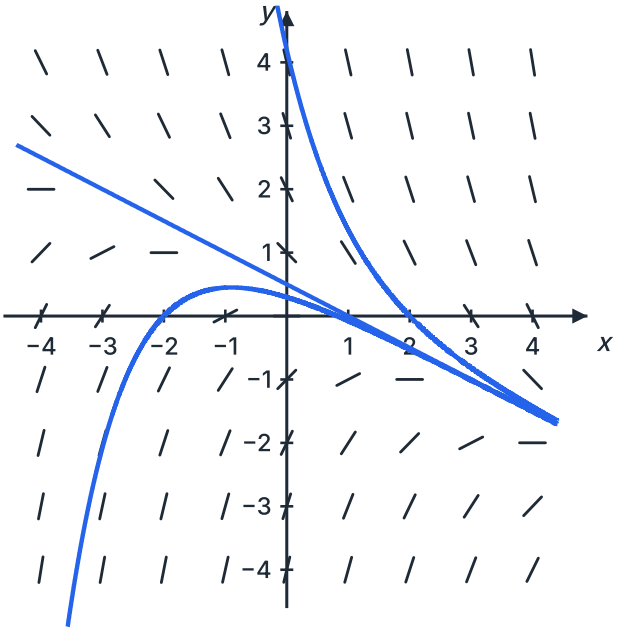
1 **a** -1 **b** 1 **c** 3 **d** $\frac{1}{2}$ **e** $\frac{1}{2}$ **f** -3

2 **a** Every row is the same: $-\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}$ for $x = -1, 0, \ldots, 5$. **c** Every entry in that column is the same. **d** concave up **e** the parabola $y = \frac{1}{4}x^2 - x$ **f** $y = \frac{1}{4}x^2 - x + 2$ and $y = \frac{1}{4}x^2 - x + 3$, vertical translations of the first curve.



3 **a** $y' = x - y$ **c** Equal entries lie along the lines $y = x - c$. **d** concave up **f** $y = x - 1$. Yes: $y' = 1$ and $x - y = x - (x - 1) = 1$.

4 **b** (i) decrease (ii) converge **c** On the line, $y' = -\frac{1}{2}x - \left(\frac{1}{2} - \frac{1}{2}x\right) = -\frac{1}{2}$, the same at every point. **d** The line has gradient $-\frac{1}{2} = y'$, so it is a solution. **e** Both solution curves approach the line as x increases: it is an asymptote.



5 **b** $y = 3$ and $y = -3$ **c** Yes, they are horizontal isoclines of horizontal line elements. **d** $y = \dfrac{3\left(e^{\frac{2}{3}x} - 1\right)}{e^{\frac{2}{3}x} + 1}$, an increasing curve through the origin, asymptotic to $y = -3$ on the left and $y = 3$ on the right, with point symmetry in the origin.

6 **a** $y = \dfrac{4}{x}$ gives $y' = -\dfrac{4}{x^2} = -\dfrac{y}{x}$, and both points lie on it. **b** $y' = x$ and $\dfrac{2y}{x} = \dfrac{x^2}{x} = x$, and $(2, 2)$ lies on it. **c** $2x - 2yy' = 0$ gives $y' = \dfrac{x}{y}$; $2x + 4yy' = 0$ gives $y' = -\dfrac{x}{2y}$. **d** At an intersection the product of the gradients is $\left(-\dfrac{y}{x}\right)\left(\dfrac{x}{y}\right) = -1$.

7 **a** $y' = c$ means $f(x) = c$, that is $x = f^{-1}(c)$: vertical lines. **b** $g(y) = c$ gives $y = g^{-1}(c)$: horizontal lines. **c** Since $y = cx + b$ is a solution, at every point on it the line element has gradient $y' = c$. So the line joins points of equal gradient: it is an isocline.

Chapter 13 Differential Equations - Homework Answers

Lesson 13B | Slope fields

Past-Paper Homework - Worked Solutions

PP 1 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q2**

The line elements are the same along each horizontal line, so y' depends on y only (this rules out A and C). They are positive above the x-axis and negative below, getting steeper as $|y|$ increases. So $y' = \dfrac{y}{2}$. **Answer D.**

PP 2 **Gosford High School, 2022, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3**

The line elements are horizontal near $(1, 1)$ and $(-1, -1)$. B and C give ± 2 at $(1, 1)$, so they are out; A and D both give 0 there. Now test $(1, -1)$, where the field slopes downwards: A gives $-1 - 1 = -2$ ✓, but D gives $1 + 1 = 2$. (Similarly at $(2, 1)$ the slope is positive: A gives $2 - 1 = 1$.) **Answer A.**

PP 3 **Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q7**

The field has both signs, which rules out B and D (they never change sign). The slopes are zero along $y = \pm x$. On the y-axis above the origin the slopes are negative, and C gives $-\frac{1}{4}y^2 < 0$ there while A gives a positive value. On the x-axis the slopes are positive, matching C ($\frac{1}{4}x^2$). **Answer C.**

PP 4 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q7**

The slopes are positive in quadrants 1 and 3 and negative in quadrants 2 and 4, so y' has the sign of xy. They are zero on both axes and steeper further out. B has the sign of y, C the sign of x, and D is never negative. **Answer (A)**, $y' = xy$.

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q9**

Only B, C and D are undefined on both axes (A is undefined only on $y = -x$). The field has the sign of xy in each quadrant and no zero slopes. D is zero on $y = \pm x$, and C, $\dfrac{x + y}{xy}$, is zero on $y = -x$. B, $\dfrac{x^2 + y^2}{xy}$, has the sign of xy and is never zero. **Answer B.**

PP 6 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q7**

The line elements are vertical along the y-axis, so the denominator is x: C or D. The slopes are horizontal along $y = -x$ (for example near $(1, -1)$), so the numerator is $x + y$. **Answer D.**

PP 7 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q12(a)**

At $A(0,-1)$: $\dfrac{0-(-1)}{1+0}=1$, so draw a short segment at 45° (gradient 1).

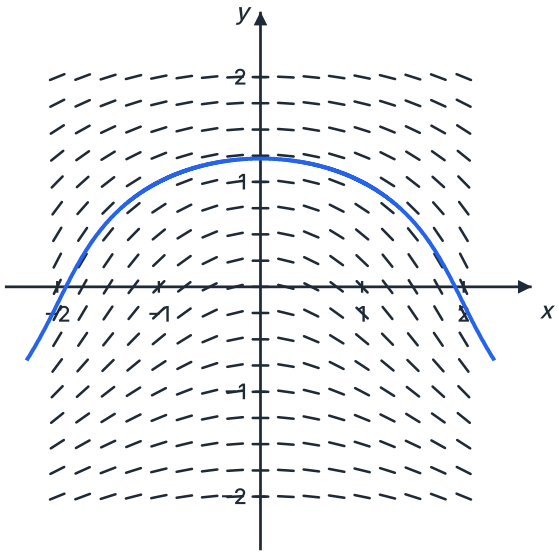
At $B(1,-1)$: $\dfrac{2-(-1)}{1+1}=\dfrac{3}{2}$, so draw a short segment slightly steeper than 45°, rising to the right.

PP 8 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(a)**

$A(0,-2)$: $\dfrac{0-2}{0+8}=-\frac{1}{4}$, a gentle downward slope. $B(-1,2)$: $\dfrac{-2+2}{-1-8}=0$, a horizontal line element. $C(1,1)$: the denominator is $1-1=0$, so the gradient is undefined and the line element is vertical.

PP 9 **Sydney Technical High School, 2020, Trial Higher School Certificate Examination (Sample 3), Mathematics Extension 1, Q14(d)**

(i) Starting at $(-1,1)$, the curve rises to a maximum on the y-axis (where $y'=0$), then falls, crossing the x-axis at about $x\approx 1.9$.



(ii) Separating: $(1+y^2)\,dy=-x\,dx$, so $y+\frac{1}{3}y^3=-\frac{1}{2}x^2+C$. At $(-1,1)$: $\frac{4}{3}=-\frac{1}{2}+C$, so $C=\frac{11}{6}$. Multiplying by 6: $6y+2y^3=-3x^2+11$, that is $2y^3+6y+3x^2-11=0$.

(Check of (i): when $y=0$, $x^2=\frac{11}{3}$, so $x\approx 1.91$.)

Chapter 13 Differential Equations - Homework Answers

Lesson 13C | Separable differential equations

Lesson Examples

C1 **Tutor demo.** $e^y\,dy = 3x^2\,dx$, $e^y = x^3 + C$, $y = \log(x^3 + C)$. Through $(0, 0)$: $C = 1$, so $y = \log(x^3 + 1)$.
Your turn. $y\,dy = 2x\,dx$, $y^2 = 2x^2 + D$. Through $(1, 3)$: $D = 7$, so $y^2 = 2x^2 + 7$ (the branch $y = \sqrt{2x^2 + 7}$).

C2 **Tutor demo.** $y = 0$ is a solution. Otherwise $y^{-2}\,dy = 2x\,dx$, so $-\dfrac{1}{y} = x^2 + C$. (i) $y(1) = -1$: $1 = 1 + C$, so $C = 0$ and $y = -\dfrac{1}{x^2}$. (ii) $y = 0$.
Your turn. $y = 0$ is a solution. Otherwise $-\dfrac{1}{y} = -\dfrac{1}{x} + C$. (i) $C = -1$, so $\dfrac{1}{y} = \dfrac{1}{x} + 1$ and $y = \dfrac{x}{1 + x}$. (ii) $y = 0$.

C3 **Tutor demo.** $y = 3$ is a solution. Otherwise $-\log|3 - y| = x^2 + C$, giving $y = 3 + Ae^{-x^2}$ for any A. (i) $A = -3$: $y = 3 - 3e^{-x^2}$. (ii) $A = 0$: $y = 3$.
Your turn. $y = 0$ is a solution. Otherwise $\log|y| = \sin x + C$, so $y = Ae^{\sin x}$. With $y(0) = 2$: $y = 2e^{\sin x}$.

Textbook Homework

1 **a** $\displaystyle\int (y + 1)\,dy = \int (x - 1)\,dx$ **b** $(y + 1)^2 = (x - 1)^2 + D$

2 **a** $y = \log\left|\frac{1}{2}x^2 + C\right|$ **b** $y = \tan(x^4 + C)$

3 **a** Substitute $y = 0$, where $x \neq 0$: LHS and RHS are both zero. **b** $y = \dfrac{1}{C + \log|x|}$

4 **a** $\displaystyle\int y\,dy = \int -x\,dx$ **b** $y^2 = -x^2 + D$ **c** $y^2 + x^2 = 4$

5 **a** $y^2 = x^2 + 1$ **b** $y = \tan\left(\frac{1}{2}(x + 1)^2\right)$ **c** $y = \dfrac{1}{x^2 + 1}$ **d** $y = \log|1 + \tan x|$

6 **a** $y = -2$ **b** $y = Cx^2 - 2$, where $C \neq 0$ **c** Allow $C = 0$.

7 **a** $y = 0$ **b** $y = Ce^{-\frac{1}{2}x^2}$, where $C \neq 0$ **c** Allow $C = 0$.

8 **a** $y = \dfrac{C}{x} + 2$ **b** $y^2 = C(1 + x^2)$ **c** $y = \dfrac{C}{x^2}$ **d** $y = Ce^{-\cos x}$ **e** $y = Ce^{-\frac{3}{x}}$ **f** $y = Cxe^{-x}$

9 **a** $\cos y = 0$, that is, $y = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}$ and $\frac{3\pi}{2}$ **b** $y = \tan^{-1}(x^3 + C)$

10 **a** $y = 0$ **b** $y = C(x-1)^2$ **c** $y = (x-1)^2$

11 **a** $y = 1$ **b** $y = 1 + C\sec x$ **c** $y = 1 + \sqrt{2}\sec x$

12 **a** $y = \frac{1}{2}x$ **b** $y = 2\sqrt{x}$ **c** $y = \dfrac{4}{1+x^2}$ **d** $y = \dfrac{2}{x}$ **e** $y = e^{\sin x - 1}$ **f** $y = e^{1-\frac{2}{x}}$

13 **a** $\dfrac{1}{x\log x}$ **b** $y = C\log x$

14 **b** $y = \dfrac{4e^x}{(2+x)^2}$

15 **a** $1 + \cos 2x$ **b** $y^2 = \sin 2x + 2x + 2$

16 **a** $y' = u + xu'$ **b** (i) $u' = \dfrac{2+u}{x}$ (ii) $u = Cx - 2$ (iii) $y = Cx^2 - 2x$

17 **a** $\tan^{-1} y + \tan^{-1} x = C$ **b** Take the tangent of both sides and use the compound-angle formula. **c** $y = \dfrac{1-x}{1+x}$

Chapter 13 Differential Equations - Homework Answers

Lesson 13C | Separable differential equations

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(b)**

$e^y\,dy = 2x\,dx \quad\Longrightarrow\quad e^y = x^2 + C$

$y(0) = \ln 4$: $4 = 0 + C$, so $C = 4$ and $y = \ln(x^2 + 4)$.

PP 2 **Abbotsleigh, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q11(f)**

$y = 0$ is a solution, but it fails $y(0) = 2$. Otherwise

$$\int \frac{dy}{y} = \int \frac{dx}{1+x^2} \quad\Longrightarrow\quad \ln|y| = \tan^{-1} x + C \quad\Longrightarrow\quad y = Ae^{\tan^{-1} x}.$$

$y(0) = 2$: $A = 2$, so $y = 2e^{\tan^{-1} x}$.

PP 3 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(b)**

$y\,dy = (1-x)\,dx \quad\Longrightarrow\quad \frac{1}{2}y^2 = x - \frac{1}{2}x^2 + C$

At $(2,0)$: $0 = 2 - 2 + C$, so $C = 0$. Hence $y^2 = 2x - x^2$, that is, $(x-1)^2 + y^2 = 1$: the circle with centre $(1,0)$ and radius 1.

PP 4 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(e)**

$$\int \frac{dy}{y^2+4} = \int x\,dx \quad\Longrightarrow\quad \frac{1}{2}\tan^{-1}\frac{y}{2} = \frac{1}{2}x^2 + C$$

At $(0,2)$: $\frac{1}{2} \times \frac{\pi}{4} = C$, so $\tan^{-1}\dfrac{y}{2} = x^2 + \dfrac{\pi}{4}$ and

$$y = 2\tan\left(x^2 + \frac{\pi}{4}\right).$$

(The solution is valid while $x^2 + \frac{\pi}{4} < \frac{\pi}{2}$, that is, $|x| < \frac{\sqrt{\pi}}{2}$.)

PP 5 **Sydney Technical High School, 2023, Trial HSC, Extension 1 Mathematics, Q13(a)**

$y = -2$ is a constant solution, but it fails $y(0) = 7$. Otherwise

$$\int \frac{dy}{y+2} = \int 3x^{\frac{1}{2}}\,dx \quad\Longrightarrow\quad \ln|y+2| = 2x^{\frac{3}{2}} + C.$$

$y(0) = 7$: $\ln 9 = C$. Since $y + 2 = 9 > 0$ initially, $y + 2 = 9e^{2x^{\frac{3}{2}}}$, so $y = 9e^{2x\sqrt{x}} - 2$.

PP 6 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)**

$$\int (2-y)\,dy = \int \frac{dx}{\sqrt{2-x^2}} \quad \Longrightarrow \quad 2y - \tfrac{1}{2}y^2 = \sin^{-1}\frac{x}{\sqrt{2}} + C$$

When $x = 1, y = 0$: $0 = \frac{\pi}{4} + C$, so $C = -\frac{\pi}{4}$. Multiplying by -2:

$$y^2 - 4y + 2\sin^{-1}\frac{x}{\sqrt{2}} - \frac{\pi}{2} = 0 \quad \Longrightarrow \quad y = 2 \pm \sqrt{4 - 2\sin^{-1}\frac{x}{\sqrt{2}} + \frac{\pi}{2}}.$$

At $x = 1$ the square root is $\sqrt{4} = 2$, and $y = 0$ needs the negative sign:

$$y = 2 - \sqrt{4 + \frac{\pi}{2} - 2\sin^{-1}\frac{x}{\sqrt{2}}}.$$

Chapter 13 Differential Equations - Homework Answers

Lesson 13D | y′ = g(y) and the logistic equation

Lesson Examples

C1 **Tutor demo.** Separating: $e^{-2y}\,dy = dx$, $-\frac{1}{2}e^{-2y} = x + C$. Reciprocal: $\dfrac{dx}{dy} = e^{-2y}$, $x = -\frac{1}{2}e^{-2y} + C$. Either way $y = -\frac{1}{2}\log(D - 2x)$, for some constant D.

C2 **Tutor demo.** $y = Ae^{2x}$; $10 = Ae^{2}$, so $y = 10e^{2(x-1)}$. **Your turn.** $y = 4$ is the constant solution; $y = 4 + Ae^{-x}$; $y(0) = 1$ gives $y = 4 - 3e^{-x}$.

C3 **Tutor demo.** $\frac{1}{2} = \dfrac{1}{1+B}$, so $B = 1$ and $y = \dfrac{1}{1+e^{-x}}$. Domain all real x, range $0 < y < 1$, horizontal asymptotes $y = 0$ (left) and $y = 1$ (right). The population starts very small, grows at an increasing rate and then a decreasing rate, approaching the stable population 1.

Your turn. $2 = \dfrac{1}{1+B}$, so $B = -\frac{1}{2}$ and $y = \dfrac{1}{1-\frac{1}{2}e^{-x}}$. The vertical asymptote is where $e^{-x} = 2$, that is $x = -\log 2$. Domain $x > -\log 2$, range $y > 1$; the curve decreases towards $y = 1$.

C4 **Your turn.** $y' = 6y - 2y^2$, so $y'' = (6 - 4y)\,y' = 2y(3 - y)(6 - 4y)$. It changes sign at $y = \frac{3}{2}$: the inflection is at height $y = \frac{3}{2}$.

Textbook Homework

1 **a** $y = 0$ **b** $\log|y| = -x + C$ **c** $y = Ae^{-x}$ **d** put $A = 0$ **e** $y = 2e^{-x}$

2 **a** $y = 0$ **b** $\dfrac{dx}{dy} = \dfrac{1}{3y}$ **c** $x = \frac{1}{3}\log|y| + C$ **d** $y = Ae^{3x}$, where A is a real number **e** put $A = 0$ **f** $y = -e^{3x}$

3 **a** $y = -3e^{x}$ **b** $y = e^{-2x}$ **c** $y = 2e^{-3x}$ **d** $y = -e^{2x}$

4 **a** $y = 2$ **b** $\log|y - 2| = -x + C$ **c** $y = 2 + Ae^{-x}$ **d** put $A = 0$ **e** $y = 2 + e^{-x}$

5 **a** $y = 1 + 2e^{-x}$ **b** $y = 1 - e^{x}$ **c** $y = -1 + 2e^{\frac{1}{2}x}$ **d** $y = 3 + e^{-2x}$

6 **a** $y = \dfrac{3}{1-6x}$ **b** $y = \dfrac{1}{1+x}$ **c** $y = \tan x$ **d** $y = -\log(1+x)$

7 **a** $y = Ae^{kx}$ **b** $y = 20e^{kx}$ **c** $k = -\log 2$ **d** $y = 20 \times 2^{-x}$, $y(3) = 2\frac{1}{2}$

8 **a** $y = Ae^{kx}$ **b** $y = 8e^{kx}$ **c** $k = \log\frac{3}{2}$ **d** $y = 8 \times \left(\frac{3}{2}\right)^{x}$, $y(4) = 40\frac{1}{2}$

9 **a** $y = \log(x + C)$ **b** Log curves with different x-intercepts. **c** (iii) Shift left or right. (iv) The isoclines are horizontal lines. **d** $y = \log(x + e)$

10 **a** $\left(0, \frac{1}{2}\right)$ **b** $e^{-x} > 0$ for all x **c** 1 and 0 **d** $L' = \dfrac{e^{-x}}{(1+e^{-x})^2}$, so $L' \neq 0$ **e** (ii) $L'' = \dfrac{e^{-\frac{x}{2}} - e^{\frac{x}{2}}}{\left(e^{\frac{x}{2}} + e^{-\frac{x}{2}}\right)^3}$ (iii) $\left(0, \frac{1}{2}\right)$ **f** an S-shaped curve rising from $y = 0$ to $y = 1$, as in the Concept 3 figure

11 **b** (i) $y = 0$ and $y = 1$ (ii) $y = \dfrac{1}{1 + Be^{-x}}$ **c** a shift of $\log B$ to the right **d** (i) yes: $y = 0$ (ii) yes: $y = 1$

12 **a** $y = 0$ and $y = 1$ **b** $y = \dfrac{1}{1 + Be^{-rx}}$ **c** $B = \dfrac{1}{y_0} - 1$ **d** (i) $B \to \infty$, so $y = 0$ in the limit. (ii) $B \to 0^+$, so $y = 1$ in the limit.

13 **b** (i) $y = 1$ and $y = 3$ (ii) $y = \dfrac{3 + Be^{-2x}}{1 + Be^{-2x}}$ **c** (i) $y = 3$ (ii) $y = 1$ **d** (i) $y'' = -2(1-y)(2-y)(3-y)$ (ii) $(0, 2)$

14 **a** $\cos^{-1} y = x + C$ **b** $C = 0$, so $\cos^{-1} y = x$ **c** LHS $= -\sin x$, RHS $= -\sqrt{1 - \cos^2 x} = -|\sin x|$. These are unequal when $\sin x < 0$. **d** $y = \cos x$, with domain $0 \leq x \leq \pi$.

Chapter 13 Differential Equations - Homework Answers

Lesson 13D | y′ = g(y) and the logistic equation

Past-Paper Homework - Worked Solutions

PP 1 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q8**

$N = 200$ is a constant solution; otherwise $\dfrac{dN}{N-200} = 0.1\,dt$, giving $N - 200 = Ae^{0.1t}$, that is $N = 200 + Ae^{0.1t}$. Only (C) has this form ($A = 100$). **Answer (C).**

PP 2 **Hurlstone Agricultural High School, 2021, Higher School Certificate Assessment Task 4, Mathematics Extension 1, Q13(c)**

$$\int \sec^2 y\,dy = \int dx \quad \Longrightarrow \quad \tan y = x + C$$

When $x = 0$, $y = \frac{\pi}{4}$: $C = 1$. So $\tan y = x + 1$ and $y = \tan^{-1}(x+1)$.

PP 3 **Gosford High School, 2023, Trial HSC Examination, Mathematics Extension 1, Q13(b)**

$\dfrac{dy}{dx} = \dfrac{3+2y}{3}$, so $\displaystyle\int \frac{dy}{3+2y} = \int \frac{dx}{3}$, giving $\frac{1}{2}\ln|3+2y| = \frac{1}{3}x + C$.

At $x = 2$, $y = -1$: $3 + 2y = 1$, so $0 = \frac{2}{3} + C$ and $C = -\frac{2}{3}$. Hence $\ln|3+2y| = \dfrac{2(x-2)}{3}$. Since $3 + 2y = 1 > 0$ at the initial point,

$$3 + 2y = e^{\frac{2(x-2)}{3}} \quad \Longrightarrow \quad y = \frac{e^{\frac{2(x-2)}{3}} - 3}{2}.$$

PP 4 **PLC, 2023, HSC Trial Examination, Mathematics Extension 1, Q14(a)**

(i) RHS $= \dfrac{1}{50} \times \dfrac{(50-A)+A}{A(50-A)} = \dfrac{1}{A(50-A)} =$ LHS.

(ii) Separating and using (i):

$$\int \frac{1}{50}\left(\frac{1}{A} + \frac{1}{50-A}\right)dA = \int \frac{dt}{25} \quad \Longrightarrow \quad \ln\left|\frac{A}{50-A}\right| = 2t + C.$$

At $t = 0$, $A = \frac{1}{2}$: $\dfrac{1/2}{99/2} = \dfrac{1}{99}$, so $C = \ln\frac{1}{99}$. With $0 < A < 50$: $\dfrac{A}{50-A} = \dfrac{e^{2t}}{99}$, so $99A = 50e^{2t} - Ae^{2t}$, and

$$A = \frac{50e^{2t}}{99 + e^{2t}} = \frac{50}{1 + 99e^{-2t}}.$$

(iii) As $t \to \infty$, $A \to 50$ cm².

PP 5 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(b)**

$\dfrac{dP}{dt} = \dfrac{0.05}{C}P(C-P)$. Separating with the given identity:

$\displaystyle\int\left(\frac{1}{P}+\frac{1}{C-P}\right)dP = \int 0.05\,dt$, so $\ln\dfrac{P}{C-P} = 0.05t + D$ (for $0 < P < C$), and

$$P = \frac{C}{1+Be^{-0.05t}}, \qquad \text{where } B = \frac{C}{50000} - 1 \text{ from } P(0) = 50000.$$

At $t = 23$: $80000\left(1+Be^{-1.15}\right) = C$. Substituting B:

$$80000 - 80000e^{-1.15} + 1.6e^{-1.15}C = C \quad\Longrightarrow\quad C = \frac{80000\left(1-e^{-1.15}\right)}{1-1.6e^{-1.15}} \approx 110\,805.$$

The carrying capacity is about 111 000.

PP 6 **Gosford High School, 2023, Trial HSC Examination, Mathematics Extension 1, Q13(c)**

(i) RHS $= \dfrac{(10000-P)+P}{P(10000-P)} = \dfrac{10000}{P(10000-P)} =$ LHS.

(ii) $\dfrac{dP}{dt} = \dfrac{0.2}{10000}P(10000-P)$, so $\displaystyle\int\frac{10000}{P(10000-P)}\,dP = \int 0.2\,dt$, giving

$$\ln\frac{P}{10000-P} = 0.2t + C.$$

At $t = 0$, $P = 1000$: $\dfrac{1000}{9000} = \dfrac{1}{9}$. So $\dfrac{P}{10000-P} = \dfrac{e^{0.2t}}{9}$, and $P = \dfrac{10000}{1+9e^{-0.2t}}$.

At $t = 7$: $P = \dfrac{10000}{1+9e^{-1.4}} \approx 3106$ rabbits.

Chapter 13 Differential Equations - Homework Answers

Lesson 13E | Applications of differential equations

Lesson Examples

C1 **Tutor demo.** a $y' + y = a - 2e^{-x} + ax + b + 2e^{-x} = ax + (a+b)$. Equating with $4x$: $a = 4$, $b = -4$. b $y'' = -3n^2 \sin nx = -48 \sin nx$, so $n = \pm 4$.

C2 **Tutor demo.** $T = 20 + 70e^{kt}$. $60 = 20 + 70e^{10k}$ gives $k = \frac{1}{10}\log\frac{4}{7} \approx -0.0560$. For 40°C: $e^{kt} = \frac{2}{7}$, so $t = \dfrac{10\log\frac{2}{7}}{\log\frac{4}{7}} \approx 22.4$ minutes.

Your turn. $M = 80e^{kt}$, with $k = \frac{1}{5}\log\frac{3}{4} \approx -0.0575$. Half-life $= \dfrac{\log 2}{-k} \approx 12.0$ days.

C3 **Your turn.** $N = \dfrac{500}{1 + Be^{-500kt}}$. $N(0) = 50$ gives $B = 9$. $N(2) = 100$ gives $e^{-1000k} = \frac{4}{9}$, so $N = \dfrac{500}{1 + 9\left(\frac{4}{9}\right)^{t/2}}$. When $t = 6$: $N = \dfrac{500}{1 + 9 \times \frac{64}{729}} \approx 279$.

Textbook Homework

1 a (i) $a = -2,\ b = 1$, so $y = x - 2x^2$ (ii) $a = -1,\ b = -1$, so $y = -e^{-x}(\cos x + \sin x)$ (iii) $a = 1,\ b = -1$, so $y = x - 1 + 3e^{-x}$
b (i) $a = \frac{1}{2},\ b = -2,\ c = -1$ (ii) $a = \frac{1}{4},\ b = \frac{5}{4},\ c = -\frac{1}{8}$ c $\lambda = -3$ or -2

2 a $R = Ae^{kt}$ b $A = 100$ c $k = -\frac{1}{4}\log 5$ d $\frac{4}{5}$ gram

3 a $H = 25 - Ae^{kt}$ b $H = 25 - 20e^{kt}$ c $k = -\frac{1}{10}\log 2$ d 43 min

4 a $h(0) = 400$; the height decreases, so $\dfrac{dh}{dt} < 0$. b $h(t) = \frac{1}{4}(kt + 40)^2$ c $k = -1$
d 40 min e No: $h(t)$ increases for $t > 40$, which is impossible.

5 a $\dfrac{y}{x}$ b $\dfrac{dy}{dx} = \dfrac{y}{x}$ c $y = Cx$ d $x = 0$, because there y' is undefined.

6 a $\dfrac{y}{x}$ b $\dfrac{dy}{dx} = -\dfrac{x}{y}$ c $x^2 + y^2 = r^2$

7 a y b $\dfrac{dy}{dx} = y$ c $y = Ae^x$ d $y = e^x$

8 a $P = P_0 e^{kh}$ b $k = -\dfrac{\log 2}{4000}$ c $160\sqrt{2} \approx 226$ kPa

9 a $A = (2x, 0)$, $B = (0, 2y)$ b $\dfrac{dy}{dx} = -\dfrac{y}{x}$ c the hyperbola $xy = C$

10 a $C = S + Ae^{-kt}$ b $C = S + (C_0 - S)e^{-kt}$

11 **b** $N = \dfrac{1000}{1 + Be^{-1000kt}}$ **c** $N = \dfrac{1000}{1 + 24e^{-1000kt}}$ **d** $k \approx 7.357 \times 10^{-4}$ **e** 623

12 **b** $N = \dfrac{P}{1 + Be^{-kPt}}$ **c** $N = \dfrac{23.2 \times 187.5}{23.2 + 164.3e^{-187.5kt}}$ **d** $k \approx 1.702 \times 10^{-4}$ **e** $N(80) \approx 120.9$ million, close to the actual 123.2 million **f** 327 million is far above the carrying capacity of 187.5 million, so the model needs revising: the carrying capacity is clearly much larger than the 1850 estimate.

13 **a** It represents the harvest (200 fish = 2 hundred per year). **b** $y(0) = 3$ **d** $y = \dfrac{12\left(1 - 3e^{-\frac{1}{3}t}\right)}{1 - 9e^{-\frac{1}{3}t}}$ **e** $t = 3\log 3 \approx 3.3$ years **f** (i) $y(0) = 5$ (ii) $y \to 12$, so the population approaches a stable 1200 fish. This suggests stopping the harvest for one year to save the species.

14 **a** $h = 20(1 + 4e^{kt})$, where $k = \frac{1}{10}\log\frac{3}{4}$ **b** (i) $\dfrac{dH}{dt} = k(H - 20 + t)$, $H(0) = 80$ (ii) $\dfrac{dy}{dt} = ky + 1$, $y(0) = 60$ (iii) $y = -\frac{1}{k} + \left(60 + \frac{1}{k}\right)e^{kt}$, so $H(t) = 20 - \frac{1}{k} - t + \left(60 + \frac{1}{k}\right)e^{kt}$

Chapter 13 Differential Equations - Homework Answers

Lesson 13E | Applications of differential equations

Past-Paper Homework - Worked Solutions

PP 1 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(b)**

(i) $R = 21$. $\dfrac{dT}{dt} = -75ke^{-kt} = -k(T-21)$, since $T - 21 = 75e^{-kt}$.

(ii) $t = 5$: $87 = 21 + 75e^{-5k}$, so $e^{-5k} = 0.88$ and $k = -\frac{1}{5}\ln 0.88 \approx 0.02557$.

$T = 70$: $e^{-kt} = \frac{49}{75}$, so $t = \dfrac{\ln \frac{75}{49}}{k} \approx 16.6$ minutes after leaving the stove.

So Miriam waits about $16.6 - 5 \approx 11.6$ more minutes (about 12 minutes).

PP 2 **Sydney Grammar School, 2023, Trial Examination, Form VI Mathematics Extension 1, Q12(b)**

(i) $\dfrac{dP}{dt} = 17ke^{-kt} = k(25 - P)$, since $25 - P = 17e^{-kt}$.

(ii) $8.072 = 25 - 17e^{-k}$, so $e^{-k} = \dfrac{16.928}{17}$ and $k = -\ln \dfrac{16.928}{17} \approx 0.00424$.

(iii) $20 = 25 - 17e^{-kt}$, so $e^{-kt} = \frac{5}{17}$ and $t = \dfrac{\ln 3.4}{k} \approx 289$ years (using $k = 0.00424$; the unrounded k gives 288).

PP 3 **Killara High School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q14(a)**

(i) Calories used per day $= 850 + 15w$. The surplus is $1600 - 850 - 15w = 750 - 15w$ calories per day, and each kilogram is 10 000 calories, so $\dfrac{dw}{dt} = \dfrac{750 - 15w}{10\,000}$.

(ii) $\displaystyle\int \frac{dw}{750 - 15w} = \int \frac{dt}{10\,000}$, so $-\frac{1}{15}\ln|750 - 15w| = \dfrac{t}{10\,000} + C$, giving $|750 - 15w| = Ae^{-0.0015t}$.

At $t = 0$, $w = 80$: $750 - 1200 = -450$, so $15w - 750 = 450e^{-0.0015t}$, and $w = 50 + 30e^{-0.0015t}$.

(iii) Yes. As $t \to \infty$, $w \to 50$ kg.

PP 4 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(b)**

(i) The volume increases by $3-2=1$ L/min, so the volume is $10+t$ L and the concentration is $\dfrac{Q}{10+t}$ kg/L. Salt leaves at $2\times\dfrac{Q}{10+t}$ kg/min and none enters, so $\dfrac{dQ}{dt}=-\dfrac{2Q}{10+t}$.

(ii) $\displaystyle\int\frac{dQ}{Q}=-\int\frac{2\,dt}{10+t}$, so $\ln Q=-2\ln(10+t)+C$ and $Q=\dfrac{A}{(10+t)^2}$.

$Q(0)=20$: $A=2000$. Then $Q(5)=\dfrac{2000}{225}\approx 8.89$ kg.

PP 5 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q13(d)**

$\displaystyle\int\frac{10}{N(10-N)}\,dN=\int\frac{10}{200}\,dt$, so $\ln\left|\dfrac{N}{10-N}\right|=\dfrac{t}{20}+C$.

At $t=0$, $N=15$: $\left|\dfrac{15}{-5}\right|=3$, so $C=\ln 3$. Since $N>10$ throughout, $\dfrac{N}{N-10}=3e^{\frac{t}{20}}$, and

$$N=\frac{30e^{\frac{t}{20}}}{3e^{\frac{t}{20}}-1}=\frac{30}{3-e^{-\frac{t}{20}}}.$$

At $t=6$: $N=\dfrac{30}{3-e^{-0.3}}\approx 13.3$, so about 13 million fish remain.

Chapter 13 Differential Equations - Homework Answers

Chapter 13 Review | Chapter review exercise

1 **a** 1st-order, linear, one arbitrary constant **b** 2nd-order, linear, two arbitrary constants **c** 3rd-order, non-linear, three arbitrary constants

3 **a** $y = 0$ **b** $\dfrac{dx}{dy} = -\dfrac{2}{y}$ **c** $x = -2\log|y| + C$ **d** $y = Ae^{-\frac{1}{2}x}$, where A is a real number **e** put $A = 0$ **f** $y = 3e^{-\frac{1}{2}x}$

5 **a** $y = \dfrac{C}{1+x^2}$ **b** $(x-1)^2 + (y+2)^2 = C$ **c** $y = \dfrac{Ce^x}{x}$

6 **a** $y = 1 + e^{-\frac{1}{2}x}$ **b** $y = 5 - 3e^{-\frac{1}{3}x}$

7 **b** $y = C\sqrt{\dfrac{1+x}{1-x}}$

8 **a** $y = \dfrac{2}{x}$ **b** $y = \log(x^2 + x - 1)$ **c** $y = \dfrac{-1}{1+2\sqrt{x}}$

9 **b** (i) $y = 0$ and $y = 1$ (ii) $y = \dfrac{1}{1+Be^{-x}}$ (iii) $y = \dfrac{1}{1+3e^{-x}}$

10 **b** (i) $y = 2$ and $y = 3$ (ii) $y = \dfrac{3-4e^{-\frac{1}{5}x}}{1-2e^{-\frac{1}{5}x}}$ (iii) $x = 5\log\frac{4}{3} \approx 1.44$